

Titanic Dataset — Exploratory Data Analysis

Internship Task 2 | Python-based EDA

Objective: Explore the cleaned Titanic dataset, calculate important statistics, identify trends, patterns and anomalies, and communicate the findings through charts and actionable insights.

1. Dataset Overview

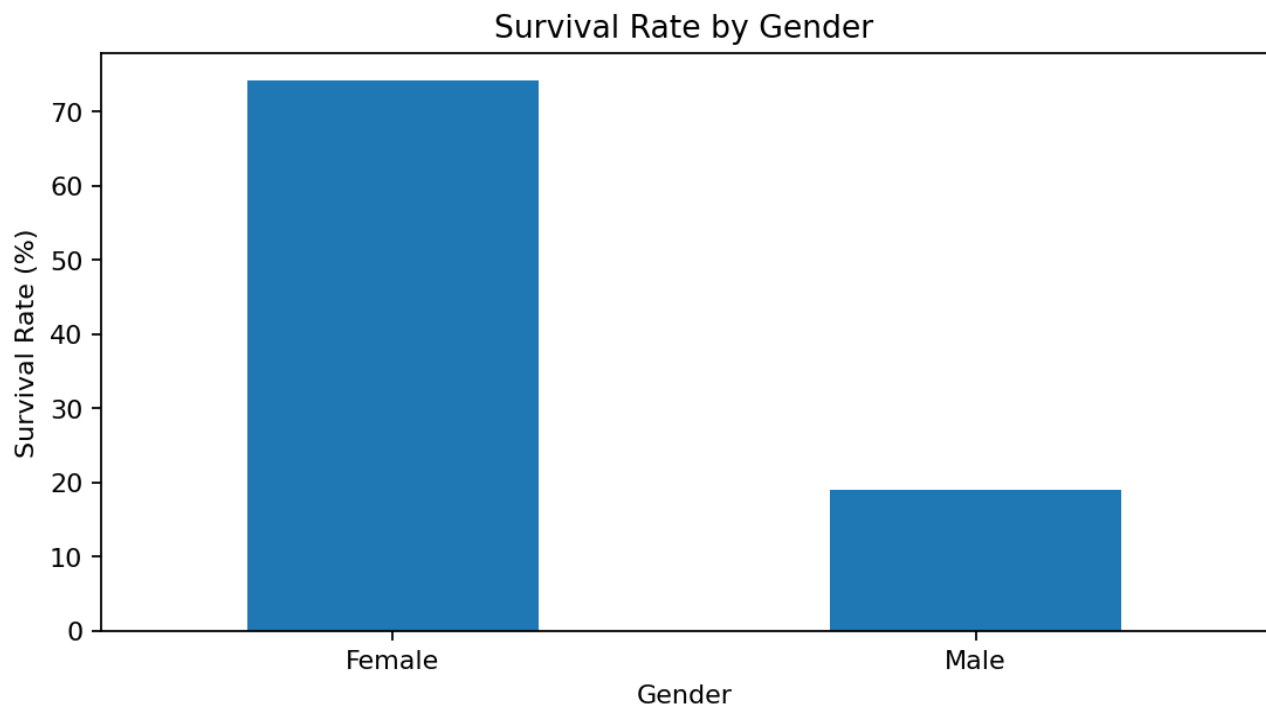
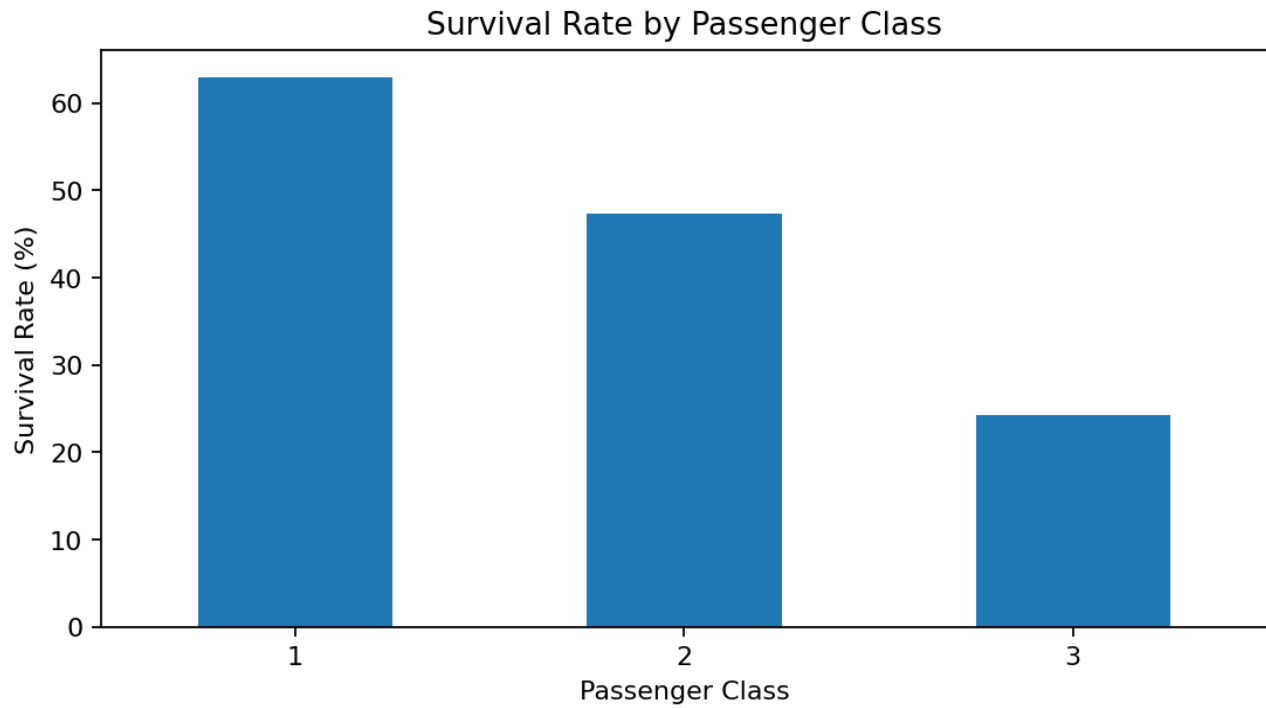
The dataset contains **891** rows and **12** columns. Duplicate records detected: **0**. Total missing cells: **0**.

Column	Data Type	Non-null	Missing
PassengerId	int64	891	0
Survived	int64	891	0
Pclass	int64	891	0
Name	object	891	0
Sex	object	891	0
Age	float64	891	0
SibSp	int64	891	0
Parch	int64	891	0
Ticket	object	891	0
Fare	float64	891	0
Cabin	object	891	0
Embarked	object	891	0

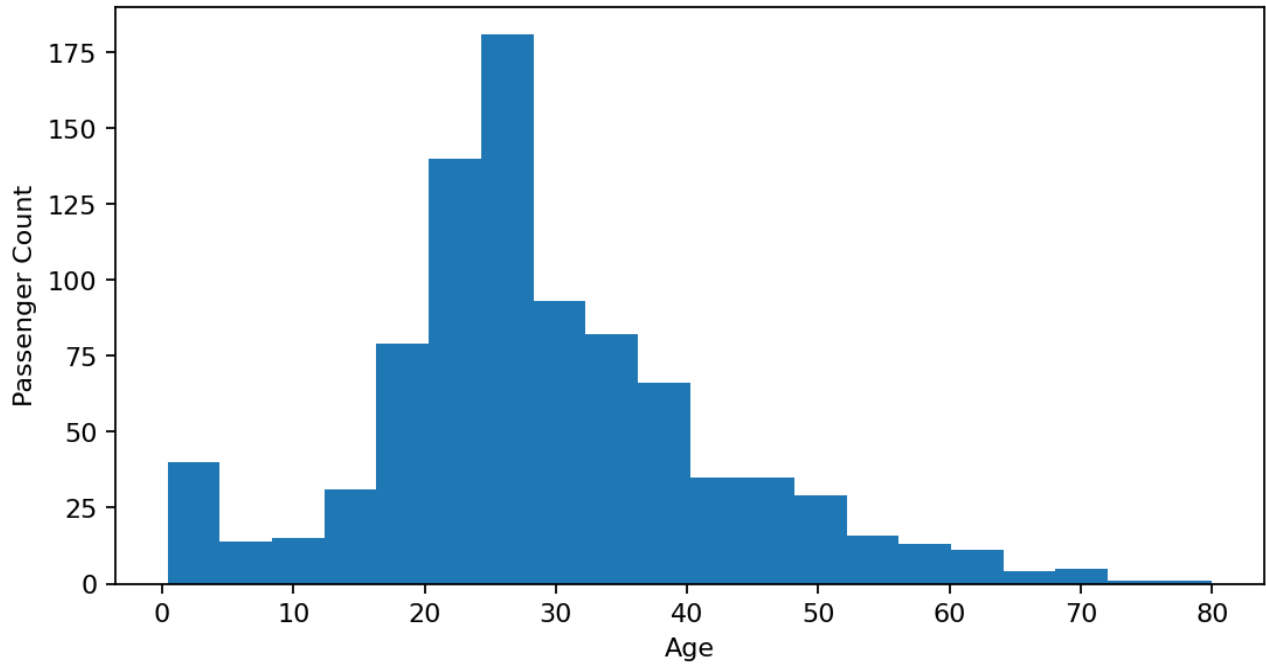
2. Descriptive Statistics

Numeric Column	Count	Mean	Median	Std. Dev.	Min	Max
PassengerId	891	446.00	446.00	257.35	1.00	891.00
Survived	891	0.38	0.00	0.49	0.00	1.00
Pclass	891	2.31	3.00	0.84	1.00	3.00
Age	891	29.11	26.00	13.30	0.42	80.00
SibSp	891	0.52	0.00	1.10	0.00	8.00
Parch	891	0.38	0.00	0.81	0.00	6.00
Fare	891	32.20	14.45	49.69	0.00	512.33

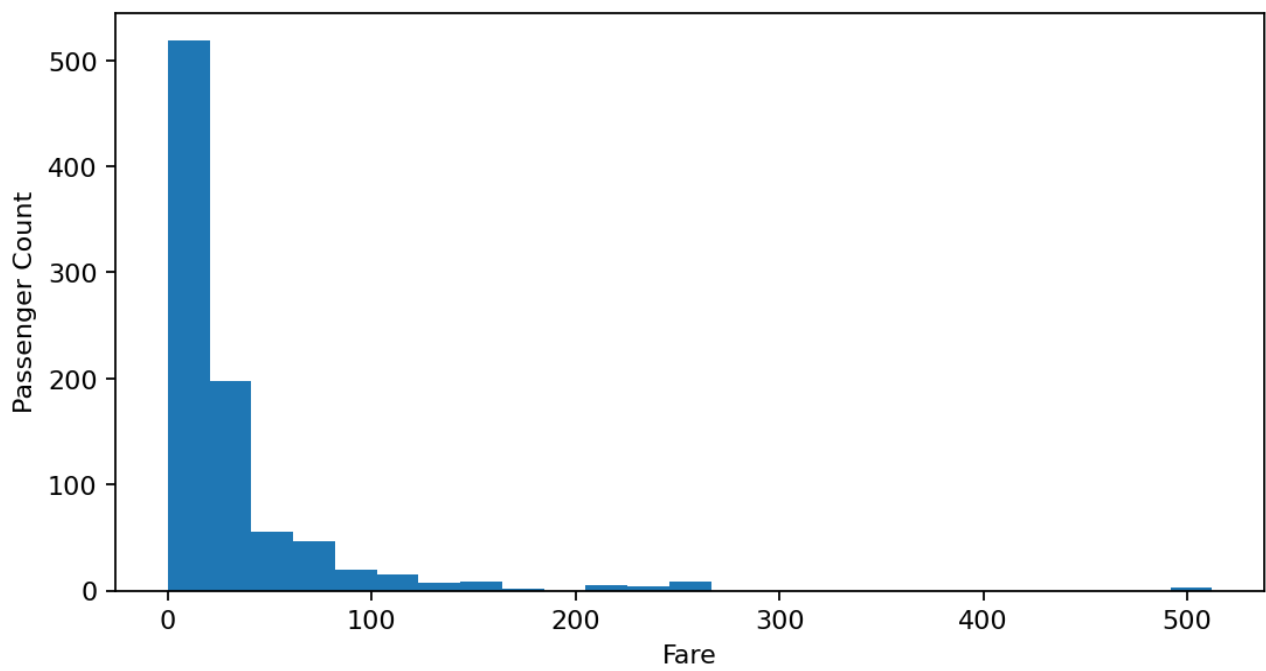
3. Visual Exploratory Analysis



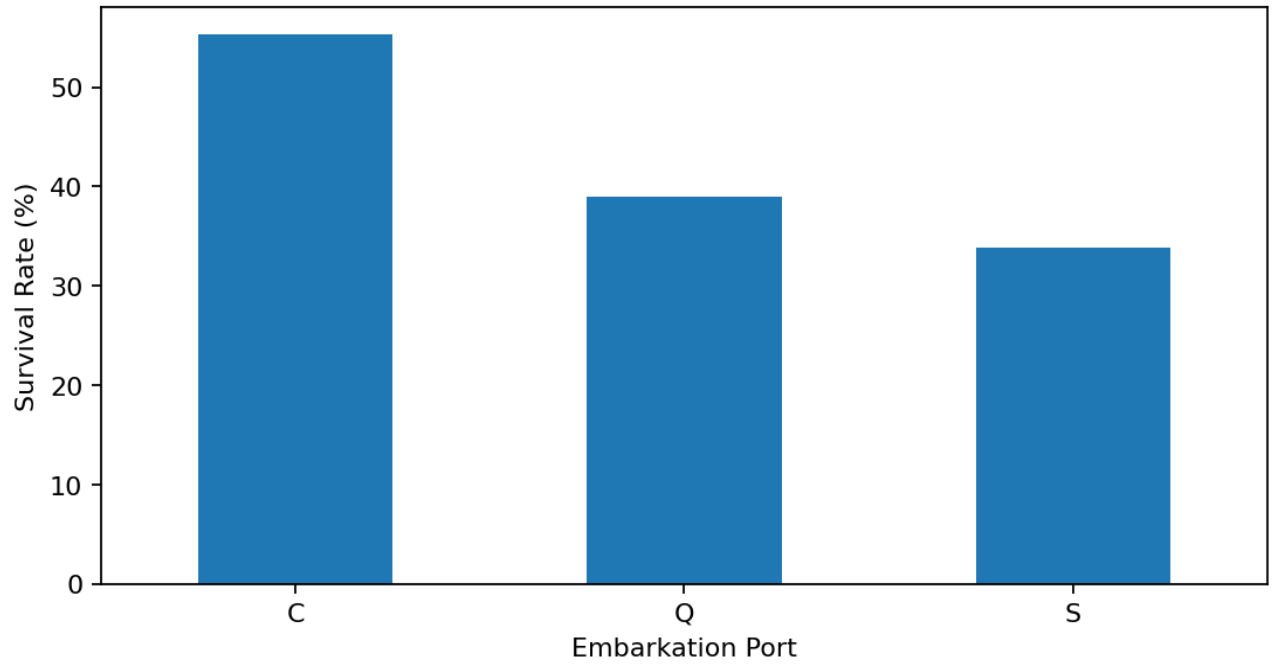
Passenger Age Distribution



Fare Distribution



Survival Rate by Embarkation Port



4. Key Insights, Trends and Patterns

Insight 1: Overall survival rate was 38.4% (342 survivors out of 891 passengers).

Insight 2: Gender shows a strong survival difference: Female had the highest survival rate at 74.2%, while Male had 18.9%.

Insight 3: Passenger class is strongly associated with survival: Class 1 had the highest rate (63.0%), while Class 3 had the lowest (24.2%).

Insight 4: Passenger age ranged from 0.4 to 80.0 years, with an average of 29.1 years and median of 26.0 years.

Insight 5: Fare values are right-skewed: the median was 14.45, while 116 fare observations were flagged as statistical outliers using the $1.5 \times \text{IQR}$ rule.

Insight 6: Survival varied across embarkation ports, with C showing the highest observed survival rate (55.4%).

5. Anomalies and Data Quality Observations

- Fare outlier thresholds using IQR: below -26.72 or above 65.63.
- Age distribution was reviewed for extreme values and overall spread; histogram shape helps identify concentration and tails.
- Duplicate-row check returned 0 duplicates.
- Missing-value check returned 0 missing cells.

6. Conclusion

The exploratory analysis shows that Titanic passenger outcomes are not evenly distributed across demographic and travel-related groups. Passenger class and gender provide particularly strong differences in survival, while age and fare distributions reveal the underlying passenger mix and potential outliers. These findings demonstrate how descriptive statistics and visualization can convert a cleaned dataset into meaningful business-style insights.

Tools: Python, Pandas, NumPy, Matplotlib, ReportLab